Science Teacher, 8B

Vasant Valley School, New Delhi

Subject: Science

3139/S8B/0023

We are pleased to present the ASSET Teacher MyBook, a comprehensive resource designed to provide you with valuable insights into your students performance and support their academic growth. This resource offers a range of features to help you better understand your students' strengths and areas for development.

Personalized Teacher Feedback: Gain tailored insights into your students' distinctive strengths and areas for development. This feedback is aimed at assisting you in providing targeted guidance to each student.
Student Performance Table: Delve into the individual question-wise performances of your students. This data-driven view will aid you in identifying specific areas where students may require additional support.
Bird's Eye View: Acquire a holistic comparison of your students' performance against the national average. This perspective will help you contextualize your students' achievements and progress.
Year on Year Analysis - Previous Batch: This report compares student performance over a year.It compares the performance of a specific class compared to the previous year.
Skill-based Summary: Explore a tabulated display of students' skill proficiency, emphasizing the skills with the highest and lowest performance. This summary enables you to gauge skill performance relative to the school and national average.
Analysis of Easy/Difficult Questions: Identify questions posing ease or challenge compared to the national norm. This section illuminates areas needing focused attention, allowing you to tailor your teaching approach accordingly.
Questions with Common Wrong Answer: This section highlights potential misconceptions through selected questions. It provides valuable insights into your students' performance within a broader nationwide context.
Practice Questions and Explanation: In the Student MyBook, students receive practice questions tailored to their needs. This Teacher MyBook also features practice questions customized for your class, complete with answers and explanations. Different practice approaches have been thoughtfully curated based on subject and skill requirements.

We recommend that you review this analysis with reference to the ASSET question paper. Your commitment to your students' growth is highly commendable, and we believe that the ASSET Teacher MyBook will further empower you in this journey.

Should you have any questions or require further assistance, please feel free to reach out to us at info@ei.study. We are dedicated to supporting your efforts in enhancing your students' learning experience.

Regards,
The ASSET Team

Kindly note: Some images may slightly appear unclear due to technical reasons, but rest assured, this is not an error.

Dear Science Teacher,

Congratulations on receiving your students' ASSET test results!

ASSET helps us understand where your students excel and where they might need more support.

Let's celebrate the strengths your students have shown in Science are:

Advanced or complex data representation or interpretation
Integrating different concepts or information for decision making

Great job nurturing these talents!

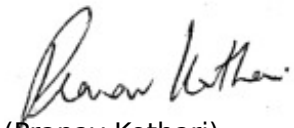
Now, let's look at areas where we can help them grow even more. Here's where they can improve in Science :

Extraction, translation and application of knowledge or information
Representing, relating or explaining scientific processes or observed phenomena

To assist you in this journey, we've created a special booklet. It's designed to focus on the questions/ skills that need a little boost. The booklet provides insights into each student's progress and highlights questions that sparked interest or posed a challenge.

Take your time going through the analysis to understand how each student is doing. You'll find different sections in the booklet to spot learning gaps and dive into specific questions that need attention. Your students have personalized booklets, called Student MyBooks, too. These are tailored to their needs, giving them a glimpse into the areas where they can practice more.

Feel free to use these resources in class, and let us know how it goes. We're here to help and learn from your experience. Share your thoughts with us at info@ei.study.As you guide your students through their learning journey, you're helping them become even better. Best of luck!



(Pranav Kothari)
Chief Executive Officer - Educational Initiatives

In case of queries, please write to us at info@ei.study or contact our customer care executive at the toll free number 1800-102-8885 (9 am - 6 pm, Mon - Fri).

No.	Student	Question No: →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	Score	Scaled Scores	Percentile	Performance
		Correct Answers →	D	B	A	D	A	B	C	D	D	B	A	C	A	C	D	D	B	C	A	A	C	D	B	B	A	B	C	B	A	D	B	D	A	A	B	A	D	B	C	C	D	C	A	D	A				
1	Aadi Lamba		C	✓	B	A	D	A	✓	✓	C	A	B	D	✓	✓	B	C	D	A	B	C	✓	✓	✓	A	D	A	✓	D	✓	B	✓	A	✓	B	✓	D	B	C	A	✓	✓	✓	✓	B	B	17	489	50	
2	Aaditva Nanda		✓	✓	B	✓	✓	A	✓	B	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	B	✓	D	✓	✓	✓	D	✓	✓	✓	✓	B	✓	✓	✓	C	✓	✓	B	A	✓	✓	✓	✓	✓	✓	D	33	708	97	
3	Abhinav Mandal		✓	✓	B	✓	C	✓	✓	B	✓	✓	B	✓	D	✓	C	C	D	✓	B	✓	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	✓	C	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	D	31	690	97	
4	Aditya Singh		C	✓	C	✓	C	✓	✓	B	✓	✓	B	D	B	✓		A		A	B	D	✓	A	✓	D	C	A	✓	D	✓	✓	✓		D	✓	A	B	C	C	✓	D	✓	✓	D	✓	B	18	517	61	
5	Aditya Tomar		✓	✓	B	✓	✓	✓	✓	B	✓	C	✓	D	✓	✓	C	C	✓	✓	B	C	✓	✓	✓	✓	B	A	B	✓	✓	C	✓	A	✓	✓	✓	✓	B	C	✓	D	✓	✓	B	B	B	26	638	92	
6	Adrit Wadhwa		✓	✓	✓	B	✓	✓	✓	A	✓	C	✓	D	✓	✓	A	C	✓	D	B	C	✓	C	✓	✓	D	✓	A	C	✓	C	✓	B	✓	✓	✓	D	C	C	B	✓	✓	✓	B	✓	✓	25	636	91	
7	Akshita Barjatiya				C	A	C		D	B	✓	A	B	A	D	✓	A	C	D	✓	✓	B	✓	✓	✓	C	✓	✓	B	D	✓	B	✓	✓	C	B	C	D	B	A	B	A	C	B	✓	✓	✓	15	450	33	
8	Ameir Sandhu		B	C	C	A	B	✓	A	C	✓	C	✓	✓	✓	D	B	✓	D	✓	D	C	A	C	D	A	B	✓	A	D	✓	✓	A	✓	✓	B	C	B	B	C	✓	D	A	A	C	C	B	13	448	32	
9	Anjani D Singh		C	✓	✓	A	D	✓	D	C	✓	✓	✓	✓	✓	✓	A	B	✓	D		D	A	✓	D	D	B	✓	✓	D	✓	C	✓	A	✓	B	C	D	B	C	A	D	A	✓	B	✓	✓	19	536	68	
10	Anoushka Barua		✓	C	C	B	✓	✓	✓	B	✓	C	✓	B	D	✓	A	C	✓	✓	B	C	D	A	D	✓	B	✓	✓	A	✓	B	✓	C	✓	✓	D	D	A	D	B	✓	B	✓	✓	✓	✓	21	553	74	
11	Aryaman Saxena		C	✓	C	A	C	✓	✓	B	✓	A	C	✓	C	✓	A	✓	✓	D	D	✓	✓	C	D	C	B	✓	A	D	✓	C	A	✓	B	C	D	D	B	A	A	D	✓	D	✓	C	✓	16	471	42	
12	Asherah J Srinivas		C	✓	C	B	D	✓	✓	C	✓	C	C	✓	✓	✓	B	C	D	✓	C	B	✓	✓	✓	D	B	✓	✓	D	✓	C	✓	C	✓	✓	A	✓	✓	C	B	B	A	✓	D	✓	D	21	575	80	
13	Bhavesh		✓	✓	C	B	✓	✓	✓	B	✓	✓	B	✓	D	B	C	A	C	✓	B	✓	D	✓	✓	✓	D	✓	✓	✓	✓	B	✓	✓	✓	B	✓	✓	C	A	✓	✓	✓	✓	B	✓	D	27	627	90	
14	Kabir Chaudhuri		✓	✓	C	A	C	✓	✓	B	C	C	✓	B	✓	✓	✓	C	✓	D	B	✓	D	✓	✓	✓	D	A	B	D	✓	✓	✓	✓	✓	C	✓	✓	B	A	✓	✓	✓	✓	✓	✓	D	26	616	88	
15	Kara Vohra		C	C	C	B	D	✓	A	B	✓	C	✓	✓	D	✓	A	B	A	A	C	D	✓	✓	C	A	B	✓	✓	D	✓	B	✓	A	✓	C	C	D	B	A	A	✓	✓	B	C	✓		15	495	52	
16	Kavya Chauhan		C	✓	C	B	C	✓	D	B	C	D	D	B	D	A	A	B	D	A	✓	D	✓	✓	✓	C	B	A	✓	A	✓	A	✓	✓	✓	✓	A	D	C	A	B	D	B	✓	✓	B	B	14	453	35	
17	Maedhaavi Mahajan		✓	✓	✓	C	C	✓	✓	A	✓	✓	✓	✓	✓	✓	A	✓	D	✓	D	C	✓	✓	C	✓	D	A	✓	✓	✓	✓	✓	A	✓	✓	✓	✓	✓	A	✓	✓	A	✓	✓	✓	✓	32	729	98	
18	Myeisha Sirohi		C	C	C	B	C	✓	A	B	C	A	✓	D	C	✓	A	C	C	D	✓	C	✓	✓	C	A	D	C	A	D	✓	✓	✓	C	✓	✓	A	B	B	A	A	✓	B	✓	B	✓	✓	15	482	47	
19	Naavya Gogoi		✓	A	B	C	D	✓	✓	C	C	A	✓	B	✓	✓	B	✓	C	D	D	C	✓	C	✓	✓	C	C	B	D	✓	C	✓	C	C	B	A	D	✓	D	A	✓	C	✓	✓	B	B	16	488	49	
20	Naintara Kapur		C	✓	C	B	C	✓	B	B	C	✓	B	D	C	✓	A	C	C	D	✓	C	D	C	A	D	✓	C			✓	C	✓	✓	✓	B	A	D	A	A	A	✓	A	B	D	✓	B	12	421	21	
21	Naisha Garg		B	C	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	C	✓	A	C	✓	D	D	B	✓	✓	C	✓	D	✓	✓	✓	✓	C	✓	✓	✓	C	✓	✓	✓	✓	A	✓	✓	B	B	✓	✓	29	638	92	
22	Natin Gujjar		C	A	C	✓	C	A	B	C	A	C	D	B	D	✓	B	A	C	D	C	B	B	A	C	D	B	A	✓	A	✓	C	A	B	D	C	A	B	A	✓	A	D	✓	✓	D	✓		8	326	1	
23	Norra K Chhatwal		C	✓	C	C	C	A	✓	A	✓	D	✓	✓	✓	✓	A	✓	✓	✓	B	D	✓	✓	C	✓	D	D	✓	✓	✓	C	✓	B	✓	B	✓	✓	A	C	B	B	✓	✓	B	✓	✓	24	593	84	
24	Raksha Joshi		C	✓	C	B	D	✓	D	B	B	D	D	B	D	A	A	C	✓	B	B	D	✓	✓	✓	D	C	✓	✓	A	C	B	✓	✓	C	✓	A	✓	✓	A	*	*	B	✓	B	C	B	14	424	23	
25	Rangita P Bachlaus		✓	✓	B	A	D	✓	✓	B	✓	✓	✓	✓	D	✓	B	✓	✓	✓	B	B	✓	✓	✓	D	B	✓	A	✓	✓	C	✓	A	✓	C	✓	✓	✓	C	B	✓	A	✓	✓	✓	✓	28	667	95	
26	Ranveer Prakash		C	C	✓	✓	✓	✓	✓	C		D	B	D	✓	✓	✓	B	A	D	D	*	✓	C	A	*	B	A	*	C	✓	✓	✓	✓	C	*	*	*	✓	*	A	✓						15	444	31	
27	Rhea Bhushan		C	A	C	B	B	✓	✓	B	C	C	✓	D	C	✓	✓	A	*	D	C	D	✓	B	C	✓	✓	C	B	D	✓	✓	C	B															10	390	11	
28	Ryan Kujur		B	C	C	A	B	✓	✓	C	✓	✓	✓	B	D	✓	B	✓	C	B	✓	B	A	C	✓	C	✓	D	B	C	B	C	D	C	D	✓	✓	C	B	A	✓	B	B	✓	D	A	C	14	427	24	
29	Shaan Kochhar		C	✓	C	C	✓	✓	✓	B	✓		✓	✓	✓	✓	B	✓	✓	✓	D	C	✓	✓	C	C	B	D	B	✓	✓	C	✓	A	✓	C	✓	D	C	C	✓	D	A	✓	C	✓	✓	23	602	86	
30	Shreyas A Mukhopadhyay		A	C	*	A	✓	A	*	B	✓	C	✓	✓	✓	✓	A	C	C		B	C	✓	✓	D	D	✓	A	✓	✓	✓	C	✓	✓	✓	B	✓	✓	C	C	B	B	C	B	B	✓	✓	19	518	61	

% answered correct (section)	32	61	16	23	29	81	68	3	68	35	61	48	45	87	10	32	42	39	16	16	71	61	52	39	19	48	48	35	94	29	84	42	71	32	48	42	29	13	32	52	42	68	32	68	39	537.4/500			
% answered correct (school students)	31	68	12	15	30	58	64	8	65	46	61	57	55	81	6	27	38	45	18	18	72	52	57	38	22	56	48	50	92	31	81	35	79	34	52	30	42	20	43	56	36	60	27	70	56	539.9/500			
% answered correct (all students)	41	59	26	44	39	58	54	12	50	41	53	43	27	75	8	35	45	44	26	27	68	50	56	35	31	48	40	37	80	32	67	28	62	36	39	34	33	27	33	40	34	51	26	55	45	505.5/500			

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. ***** Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

		Question No: →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	Score	Scaled Scores	Percentile	Performance
No.	**Student↓**	**Correct Answers →**	**D**	**B**	**A**	**D**	**A**	**B**	**C**	**D**	**D**	**B**	**A**	**C**	**A**	**C**	**D**	**D**	**B**	**C**	**A**	**A**	**C**	**D**	**B**	**B**	**A**	**B**	**C**	**B**	**A**	**D**	**B**	**D**	**A**	**A**	**B**	**A**	**D**	**B**	**C**	**C**	**D**	**C**	**A**	**D**	**A**				
31	Smita Parida		C	✓	C	A	C	✓	✓	B	✓	✓	✓	✓	C	✓	A	✓	✓	A	D	C	✓	C	✓	D	B	✓	A	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	B	✓	A	B	C	✓	D	25	609	87	

% answered correct (section)	32	61	16	23	29	81	68	3	68	35	61	48	45	87	10	32	42	39	16	16	71	61	52	39	19	48	48	35	94	29	84	42	71	32	48	42	29	13	32	52	42	68	32	68	39	537.4/500			
% answered correct (school students)	31	68	12	15	30	58	64	8	65	46	61	57	55	81	6	27	38	45	18	18	72	52	57	38	22	56	48	50	92	31	81	35	79	34	52	30	42	20	43	56	36	60	27	70	56	539.9/500			
% answered correct (all students)	41	59	26	44	39	58	54	12	50	41	53	43	27	75	8	35	45	44	26	27	68	50	56	35	31	48	40	37	80	32	67	28	62	36	39	34	33	27	33	40	34	51	26	55	45	505.5/500			

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. ***** Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

The Bird's Eye View shows a comparison of the performance of your students with the national average performance.

Your class 8 B, has performed better than the national average in the Science ASSET paper this year.

Class 8	Science	No. of students
8A	109%	33
8B (Your Class)	107%	31
8C	107%	34
All sections combined	108%	98

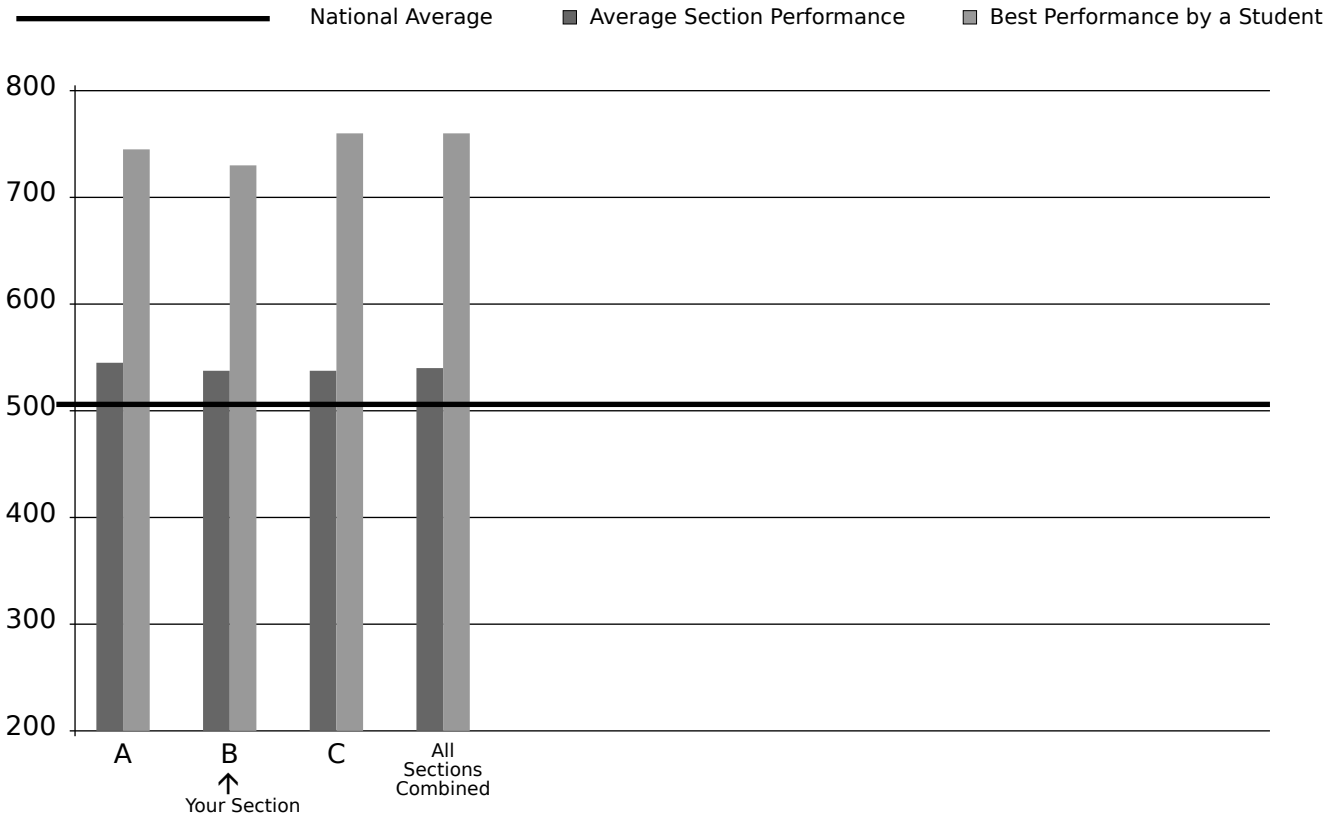
Performance as a percentage of National Average

Above Average(>105%) Below Average(<95%) Within Average(95% to 105%)

National average is taken as 100%. The value in the table indicate performance as compared to the national average.**E.g.** For a class average of 520 and overall average as 500, overall benchmarking %= (520/500)*100 = 104%. 95% to 105% is treated as similar to national average performance.Performance greater than 105% and below 95% is considered as performance above and below the national average range respectively.

Performance Data

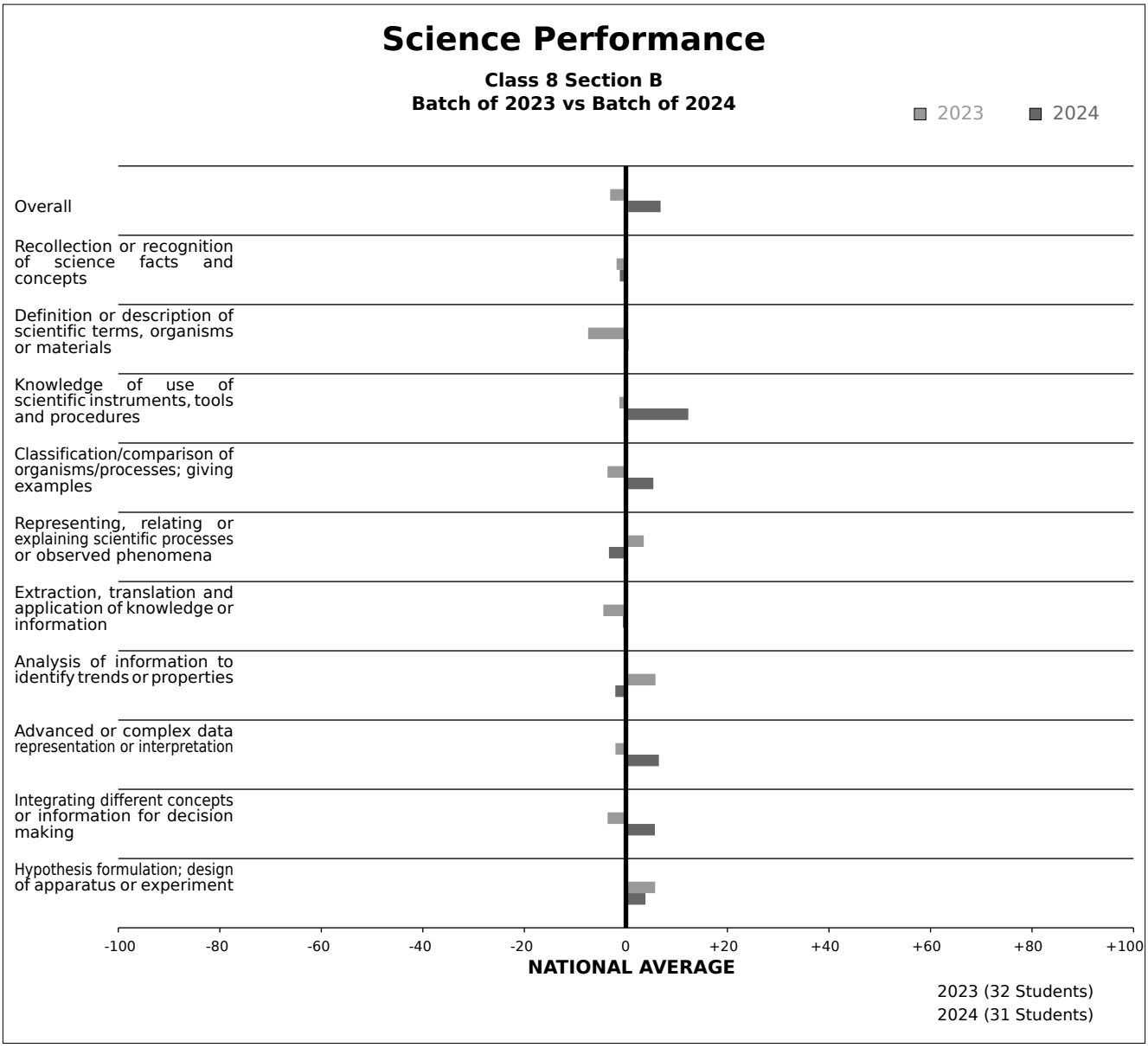
Comparison of section average and best performance with all schools in the country.



Comparison of performance of students of the current batch with the previous batch:

The following graph compares the performance of your current batch of students against the previous batch. The horizontal bars in the graph show how students of your class have fared against the national average in each skill across two years.

The thick black vertical line in the centre marks the national average.For example, a bar towards the right would mean that students performed higher than the national average that year in the particular skill.



This table shows how your students' performance compares to the overall school performance and the national average (all other schools in India). The skills with a '*' in the first column, if any, are those in which your students have performed better than the national average.

Class: 8 B Subject: Science Number of Students: 31

S.No.	Skill	Questions	Class 8 B Performance	Class 8 Performance (School)	Class 8 Performance (National)
			Average	Average	Average
1	Recollection or recognition of science facts and concepts	1,6,16,18,20	40.0%	35.7%	41.2%
2*	Definition or description of scientific terms, organisms or materials	3,17,31,38,44	44.5%	44.3%	44.0%
3*	Knowledge of use of scientific instruments, tools and procedures	7,13,22,43	51.6%	49.5%	39.3%
4*	Classification/comparison of organisms/ processes; giving examples	10,14,24,40	53.2%	55.1%	47.8%
5	Representing, relating or explaining scientific processes or observed phenomena	2,8,34	32.3%	36.7%	35.6%
6	Extraction, translation and application of knowledge or information	5,15,19,26,32,39	29.6%	31.3%	30.2%
7	Analysis of information to identify trends or properties	4,12,25,33,36	40.6%	40.6%	42.7%
8*	Advanced or complex data representation or interpretation	29,30,35	57.0%	58.2%	50.5%
9*	Integrating different concepts or information for decision making	9,21,41,45	54.8%	57.4%	49.1%
10*	Hypothesis formulation; design of apparatus or experiment	11,23,27,28,37,42	48.9%	53.1%	45.1%

Note - The average performance is calculated from the number of correct responses given by students to the questions constituting specific skills.

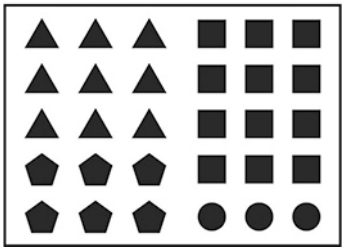
Strong skills		Weak skills	
1.	Advanced or complex data representation or interpretation	1.	Extraction, translation and application of knowledge or information
2.	Integrating different concepts or information for decision making	2.	Representing, relating or explaining scientific processes or observed

This part of the report shows questions that the children of your class found easy and difficult. This information can be used by you to focus your attention to specific questions and skills. You can also use the student performance table to focus on individual students. (Please refer to the ASSET paper for the complete text of these questions.)

Questions which students of your school found the MOST DIFFICULT compared to the rest of the country:

Q 4 Zubeda used four unique symbols to represent the students in her class as shown.

Students in class VIII



In Zubeda's class -

- 30% of the students come to the school on their cycles everyday.
- 50% of the students are girls.

Based on the given information, what does the

▲

symbol represent?

A boys who use a cycle to come to school
B girls who use a cycle to come to school
C boys who do not use a cycle to come to school
D (We cannot say for sure based on the given information.)

Skill: Analysis of information to identify trends or properties

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
4	38224	D	23%	44%	Your School	
					All Schools	

Q 38 Susan had a large container filled with substance X. She completely filled two identical containers P and Q with substance X at room temperature and then closed them. On weighing the filled containers P and Q, she found their weights to be different.

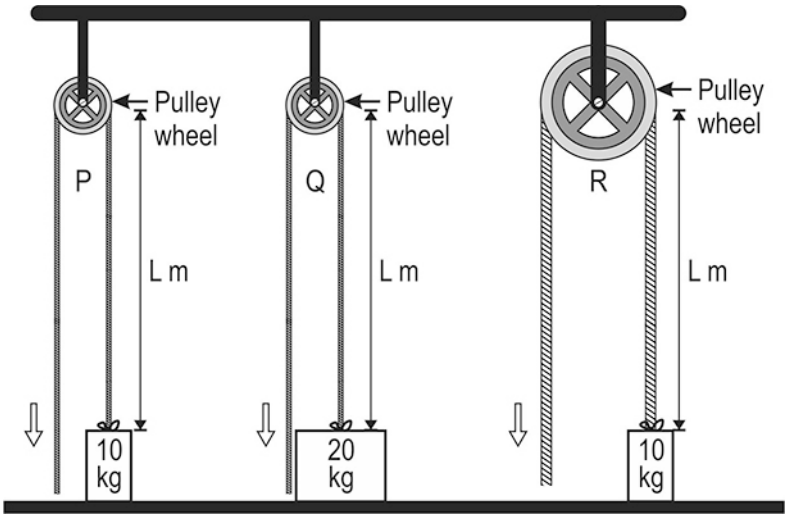
Which of the following could substance X be?

A coconut oil **B** oxygen **C** butter **D** water

Skill: Definition or description of scientific terms, organisms or materials

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
38	38224	B	13%	27%	Your School	
					All Schools	

Q 25 Pulleys P, Q and R are fixed at the same height from the ground. The diameter of pulley R is twice that of P and Q. 10 kg weights are attached to pulley P and R, while a 20 kg weight is attached to pulley Q, as shown below.



Which pulley wheels will make the SAME number of turns when the weights are pulled up the same height 'L' m?

A P and Q **B** P and R **C** all - P, Q and R **D** (none of them)

Skill: Analysis of information to identify trends or properties

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
25	38224	A	19%	31%	Your School	
					All Schools	

Questions which students of your school found the EASIEST compared to the rest of the country:

Q 6 What is the BEST time to watch vultures gliding in the sky?

A early morning **B** around midday **C** late evening **D** any time of the day

Skill: Recollection or recognition of science facts and concepts

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
6	38224	B	81%	58%	Your School	
					All Schools	

Q 13 One evening, four friends were chatting in a park.

Anil: There are so many Mango trees here.
Bhavya: I bet there are many birds in the park.
Charita: There are probably many mosquitoes here.
Danny: My doctor told me that mosquito bites can cause Malaria.

Who has made a statement that a scientist would call an OBSERVATION?

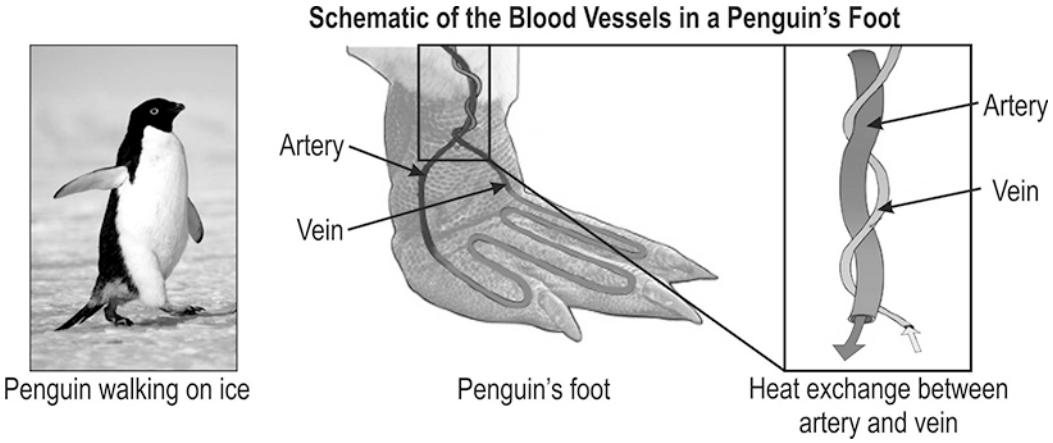
A Anil **B** Bhavya **C** Charita **D** Danny

Skill: Knowledge of use of scientific instruments, tools and procedures

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
13	38224	A	45%	27%	Your School	
					All Schools	

Q 9 Penguins in Antarctica walk on ice for their routine activities. The schematic below shows the blood vessels in the feet of penguins.

It shows the artery in close contact with the vein. This close contact allows heat to be transferred from the warmer blood flowing in the arteries to the blood flowing in the veins. Less warm blood reaches the foot through the arteries, keeping the foot cold, and thus reducing the temperature difference between the penguin's foot and the ice.



How does this type of circulation in the feet help the penguin?

A It helps it to swim faster in water.
B It helps to supply more oxygen to the feet.
C It helps to keep the feet warmer than the body.
D It helps to reduce the amount of heat lost through the feet.

Skill: Integrating different concepts or information for decision making

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
9	38224	D	68%	50%	Your School	
					All Schools	

From the questions that students across the country have found the most difficult, we have chosen a few to highlight potential misconceptions that students might have. For each question, we have given a break-up (in percentage form) of the number of students choosing a particular option. This has been done both for the section of your school and all the students nationally. Compare these to find out how your students have performed.

We have highlighted in light grey the correct option and in dark grey the most common wrong answer, and provided an explanation of why students are choosing the most common wrong option. It is recommended that you discuss these questions with the students to uncover students' rationale in choosing the most common wrong option. (Please refer to the ASSET paper for the complete text of these questions.)

Q 1 The liver secretes bile, needed to digest fat in our food. The pancreas secretes several enzymes and hormones needed to break down food.

Which of the following is true of the food that we eat?

A It passes only through our liver.
B It passes only though our pancreas.
C It passes through both our liver and pancreas.
D It passes neither through our liver nor pancreas.

This question was designed to test whether students know that food does not pass through our liver and pancreas. The most common wrong answer indicates that students feel that all the food passes through both the liver and the pancreas! The process of digestion and the path of ingested food should be discussed among students to see what other notions they may have.

Papercode: 38224, Question no: 1					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	3%	10%	52%	32%	31
National	9%	11%	38%	41%	11685

Q 15 Pollination is a step in the process of sexual reproduction in plants. It can take place in three ways.

P) A male gamete from a flower fuses with the female gamete of the same flower to form a seed.
Q) A male gamete from a flower fuses with the female gamete of another flower on the same plant to form a seed.
R) A male gamete from a flower fuses with the female gamete of a flower on another plant (of the same species) to form a seed.

Planting a seed produced by which of the methods of pollination is likely to produce a plant that has characteristics IDENTICAL to the parent plant?

A only P **B** only P and Q **C** all, P, Q and R **D** none of P, Q and R

The question tests students' understanding that ALL the three methods of pollination described involve sexual reproduction. Students choosing the correct OPTION D understand that sexual reproduction leads TO the mixing of gametes LEADING TO the formation of characteristics IN the offspring that are NOT exactly the same AS those IN the parent plant. Students choosing OPTION A might be thinking that since gametes FROM the same flower are involved, the characteristics transmitted will be the same AS the parent plant. Students choosing OPTION B might be thinking that since the flowers ON the same plant are involved, the characteristics transmitted will be the same AS the parent plant. Students choosing OPTION C might be thinking that since the flowers ON the same species are involved, the characteristics transmitted will be the same AS the parent plant.

Papercode: 38224, Question no: 15					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	52%	26%	10%	10%	31
National	28%	32%	31%	8%	11685

Q 16 Plants produce food in the form of carbohydrates in the leaves during the process of photosynthesis.

Which of the following normally happens to the food produced?

i	It is broken down into components that are used in the synthesis of proteins and fats.
ii	Extra food produced is stored as starch in different parts of the plant.
iii	It is utilised by the cells of the plant during respiration to produce energy.

A only i and ii **B** only i and iii **C** only ii and iii **D** all - i, ii and iii

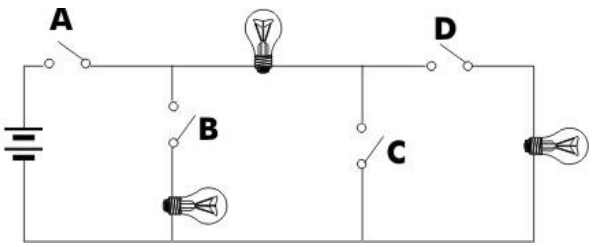
This question was designed TO test IF students understand what happens TO the food produced BY a plant. Students are familiar WITH the concept that the glucose (carbohydrate) produced BY a plant during photosynthesis IS utilized BY the cells TO produce energy BY respiration, AND that excess glucose IS converted TO starch which IS stored IN different plant parts, therefore they may choose OPTION C. Students should know that besides carbohydrates, a plant also requires proteins AND fats. They should understand that plants also synthesize other substances containing carbon, hydrogen AND oxygen LIKE proteins AND fats. The carbon, hydrogen AND oxygen used TO make proteins AND fats comes FROM the breakdown of the carbohydrate, whereas the nitrogen IN proteins comes FROM soil nutrients absorbed BY the roots.

Papercode: 38224, Question no: 16					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	13%	13%	42%	32%	31
National	17%	17%	30%	35%	11685

This section is crafted to aid students in practicing weaker skills. Use these in class discussions, gauge common understanding gaps, and tailor lessons. Answers and explanations are included. Employ a mix of question types to stimulate students' thinking. Here's how we've tailored the practice questions for each subject: For English, if reading comprehension is a challenge, we've provided full passages to work on. In Math, Science, and Social Studies, we've chosen questions based on areas where you've shown room for growth. Hindi practice focuses on non-reading comprehension skills. Computational Thinking covers a variety of skills across the class.

Skill: Extraction, translation and application of knowledge or information (30%)

Q 1 Which switch A, B, C or D MUST be open to ensure that no bulb will glow?



Explanation:

A: Look carefully at the given circuit. Let us imagine each switch being open at a time. If switch B is open, 2 bulbs will still glow. If switch C is open, all 3 bulbs will glow, because C is not needed to complete the circuit between the battery and the bulbs. If switch D is open, 2 bulbs will glow - the bulb next to B and the one before D. However, if switch A is open, the wire leading to the end of the battery will be disconnected, and hence the circuit will not receive any power. Thus, option A is the only correct answer.

Q 2 You know that unlike magnetic poles attract - the south pole of one magnet is attracted to the north pole of another magnet within its magnetic field.

A magnetic compass responds to the earth's magnetic field and comes to rest in the North-South orientation with the North pole of the compass pointing to the Earth's geographic North.

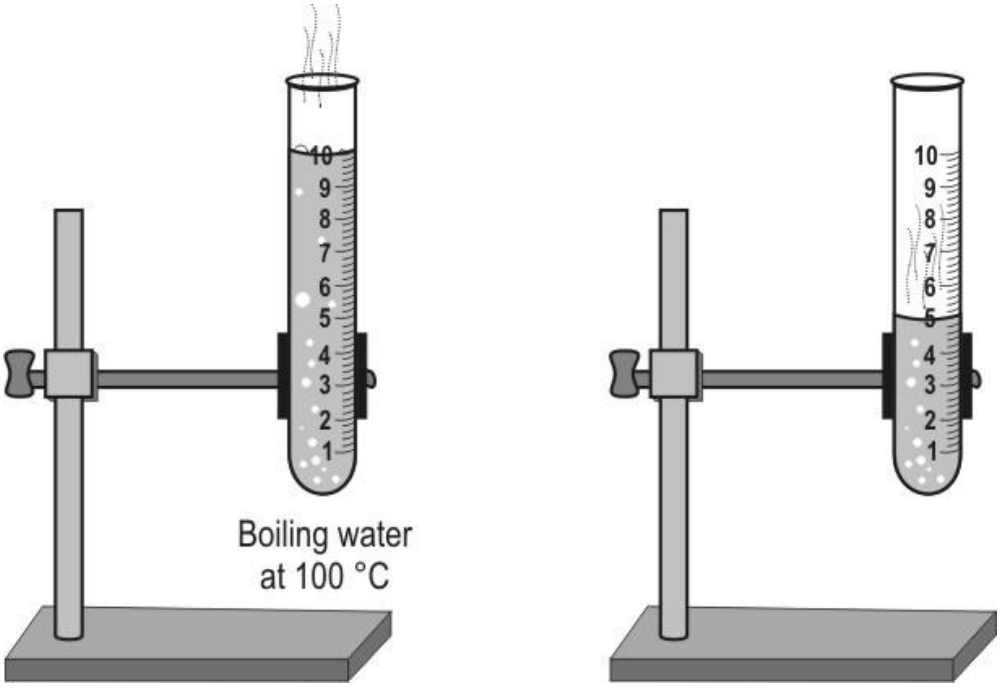
What does this imply?

A The earth's magnetic South pole is near its geographic North pole.
B The earth's magnetic North pole is near its geographic North pole.
C The earth's magnetic poles are perpendicular to its geographic poles.
D The earth's magnetic poles lie at the core of the earth, which is a mass of molten iron.

Explanation:

A: The magnetic poles of a magnet are named according to the direction they point in. The north poles of all magnets point to the North direction. We know that unlike magnetic poles attract. The reason magnets align in a certain direction on earth, is because there is a giant magnet within the earth. The South pole of the earth's magnet is towards the geographic North pole. This is the reason it can attract the North pole of other magnets. The earth's magnetic poles cannot be perpendicular to the geographic poles because the other magnets would, then, also be aligned perpendicular to the geographic poles, which does not happen. If the earth's magnetic North pole was near its geographic North pole, then the South poles of all other magnets would point to the earth's geographic North pole!

Q 3



Shakira heated 10 ml of water in a test-tube till it started boiling. She noted that the temperature of the water was 100°C. If she pours out half the water into the sink, what will be the approximate temperature of the water left in the test-tube immediately after this?

A 30°C
B 50°C
C 100°C
D 200°C

Explanation:

C: Water boils at 100°C. This is true for any amount of water. If you take boiling wate and quickly pour out some of it, and then immediately measure its temperature, it will still be 100°C. This is because heat takes some time to escape from the water. If water is boiling at 100°C, any part of the water will be 100°C, even if some water is taken out of it. Thus, only option C is correct.

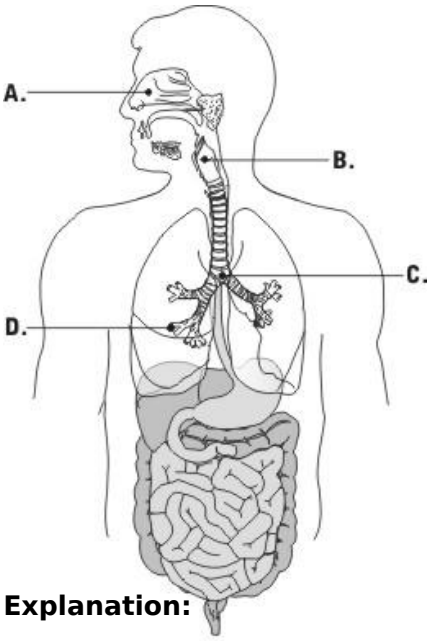
Q 4 Farmer John grew a variety of plants on his farm. For some of his plants he always preferred to use the asexual mode of reproduction rather than the sexual mode of reproduction. Which of these reasons would explain why he did this?
The resulting plants always _____.

A produced bigger and better fruits than the parent plant
B had characteristics identical to the parent plant
C produced more seeds than the parent plants
D had only the best characteristics of the parent plants

Explanation:

B: Plants that grow as a result of asexual reproduction will have characteristics identical to their parents because they grow from an older part of the parent plant, and not a new seed. They will not have bigger or better fruits, since the plant will have characteristics that are identical to the parent plant. A plant that grows from a new seed, formed due to sexual reproduction, will not be identical to the parent plants. Hence, only option B is correct.

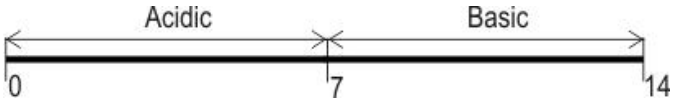
Q 5 A valve is a structure in a hollow organ that controls the direction of the flow of matter through it. Shown here is a schematic diagram of the respiratory and the digestive system. At which of the positions are you likely to find a valve?



Explanation:

B: Look at the labelled parts in the diagram. Point A is the nose, which does not require a valve. Point C is the division between the wind-pipe and the lungs, which does not require a valve either. Point D is the part where the branches of the bronchi divide into the lungs - they also, do not require a valve. Only point B, which has the openings of the respiratory and the digestive system, need a valve so that food does not enter the wind-pipe and choke us.

Q 6 Shown below is the range of pH values and its significance.



Chemical	pH
Hydrochloric acid	2
Ammonium hydroxide	10
Sulphuric acid	1
Acetic acid	5

The strongest acid among the substances listed above is

A Acetic acid
B sulphuric acid
C ammonium hydroxide
D hydrochloric acid

Explanation:

B: The pH scale in the first figure shows that acidic substances lie within the pH range between 0 to 7. Basic substances lie between 7 and 14. This means that as the pH range goes from 0 to 14, the acidity decreases and the basicity increases. The opposite is true as the pH decreases from 14 to 0. Thus, the substance with the lowest pH will be the strongest acid. In this case, it is sulphuric acid, option B.

Q 7 The human body (especially when dry) is a poor conductor of electricity, yet we get an electric shock because it does conduct some electricity. Which of these is a similar example when a poor conductor allows electricity to pass through it?

A During a thunderstorm, lightning strikes when electric charge passes through the air, which is normally a poor conductor.
B Electricity passes through the copper wires inside the motor of a ceiling fan as copper is a good conductor of electricity.
C Electricity passes through the metallic filament in a bulb when it glows, though the filament is normally a poor conductor.
D Electricity passes through the walls of a building when a light is turned on, though bricks are normally poor conductors.

Explanation:

A: Lightning is the result of a very heavy build up of opposite charges between the clouds and the earth's surface or between clouds. This build-up of charges results in a sudden discharge of electricity, even through a poor conductor like air. Copper wires and metallic filaments are not poor conductors of electricity. The electricity for lights comes through metal wires, and not directly through the bricks, hence, option D is incorrect.

Q 8 Sap is the name given to the liquids transported by xylem and phloem. What does it contain?
P. products of photosynthesis
Q. chlorophyll from the leaves

R. water and minerals from the soil
S. waste fluids excreted by the plants

A P and R **B** Q and R **C** Q and S **D** P and S

Explanation:

A: The xylem mainly transports water and minerals, and the phloem mainly transports the food produced by photosynthesis. Chlorophyll is never transported from the leaves, and the waste fluids are usually given out by the plant organs themselves. Thus, it is mainly P and R that are found in the xylem and phloem.

Skill: Representing, relating or explaining scientific processes or observed phenomena (32%)

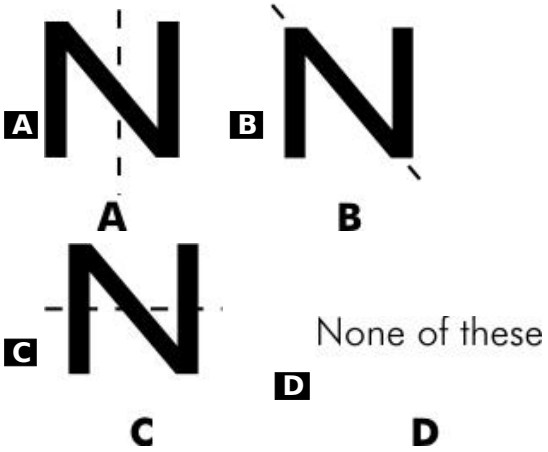
Q 1 If the temperature of inhaled air is around 24 degrees Celsius (room temperature), what would be the temperature of exhaled air?

A less than the room temperature
B equal to the room temperature
C around body temperature
D double the room temperature

Explanation:

C: The air that is inhaled goes inside the lungs where exchange of gases takes place. If the inhaled air is at a lower temperature (24 degree Celcius), heat from the body will flow to the air and make it hotter. This will happen till air and the body are at the same temperature, ie. the body temperature. So the exhaled air will be of the same temperature as that of the body. Normal body temperature is around 37 degree Celsius, so that will be the temperature of exhaled air. It won't be double the room temperature.

Q 2 A mirror is placed along the dotted line on the letter N. In which position will the image in the mirror coincide EXACTLY with the other half of the letter?



Explanation:

D: You should just draw this on a tracing paper and see in which case the two halves coincide exactly. You will find that in none of the three will the two halves coincide. Try drawing one half, place a mirror on it and then see the image that will coincide exactly with it.

Q 3 The stomach normally secretes acid that is essential in the digestive process. This acid helps in breaking down the food during digestion. When there is excess production of acid in the stomach, it results in the condition known as acidity. Antacid tablets like Digene or Gelusil are consumed by people suffering from acidity.
Which of the following CORRECTLY explains how antacids help relieve the symptoms of acidity?

A They act like sponges and absorb the acid from the stomach.
B They are powerful acids which make the stomach acids weak.
C They are bases which react with the stomach acids and lower the acidity in the stomach.
D They are neutral substances which react with the stomach acids and neutralize them.

Explanation:

C: Acids can be neutralised by adding bases to them. This simple solution is used to treat acidity in the stomach. Antacid tablets have bases which help in neutralising the acids in the stomach, and bring down the acidity. Neutral substances won't be helpful in bringing down the acidity. Only bases can neutralise acids, and not neutral substances.

Q 4 In a laboratory, a sealed container with 100 g of steam (water in its gaseous form) is cooled until all the steam becomes liquid. The container is then cooled further until all the water becomes solid. During this entire process, which of the following remains constant?

A the mass of the water
B the heat inside the container
C the pressure inside the container
D the volume occupied by the water

Explanation:

A: Since the container is sealed, no vapour or water escapes it. So the number of water particles remains the same. Hence, the mass of water will remain the same in this entire process. As steam, which is in the gaseous state, changes to water, it will occupy less volume than before. When water freezes, it will expand a little, but the volume will still be less than what the steam had occupied. So, the volume occupied by the water does change in the process.

Q 5 Water boils at 100°C. We observe that wet clothes on a clothesline lose their water and become dry even though the temperature is much less than 100°C. Which of the following explains why?

A Water has a special property by which it can "boil" even at room temperatures.
B In the presence of a flame water boils at 100°C, otherwise it can boil at any temperature.
C The material of the clothesline absorbs the water from the wet clothes and water evaporates from the clothesline.
D At any temperature, when there is water, some molecules are always escaping the surface to become water vapour.

Explanation:

D: Water changes from liquid state to its gaseous state from the surface at all temperatures. The rate at which the water molecules evaporate, depend on the temperature and the humidity (measure of how much vapour/water molecules are already present in the air). This is the reason why clothes dry up. If the material is absorbing water, it will remain wet and not dry up. For the clothes to dry up, the water molecules must escape and change to water vapour.

Q 6 A patient with cancer of the food pipe, cannot eat and swallow. So to feed him, a tube is inserted directly into his stomach. Properly mashed food is now given to the patient directly through the pipe.
Now that the food is going directly to the stomach instead of through the food pipe, which of these is likely to happen?

A water will not be absorbed
B the kidney will stop working
C carbohydrates will not be digested
D (no major problem will arise)

Explanation:

D: When we eat food, it is chewed inside the mouth and mixed with saliva. Breakdown of food happens and it passes through the foodpipe to the stomach. So what reaches the stomach is just broken down food particles along with a little bit of saliva. So, if properly mashed food is directly given to the stomach, it won't cause any major problem. Saliva helps in digesting carbohydrates, but only partly. Mainly it is digested in the small intestine. So even carbohydrate digestion will occur.

Q 7 Air contains about 21% oxygen. When we breathe in air, some of the oxygen is used and carbon dioxide is produced. What happens to the NITROGEN in the air we breathe in?

A It gets converted to proteins.
B It is also converted into CO_2.
C It comes out of the body unchanged.
D Only the O_2 in air is breathed in and CO_2 is breathed out.

Explanation:

C: We breathe in air through our nose and mouth, and not just oxygen. The air then enters the lungs where oxygen, from the air, enters the blood and carbon dioxide from the blood enters the lungs. The carbon dioxide rich air is then exhaled out. Only exchange of oxygen and carbon dioxide happens in the lungs. The other gases, including nitrogen remain unchanged.

Q 8 When we see with only one eye (by covering the other eye with our hands), what is the difference in what we see?

A There is no difference.
B We cannot distinguish colours.
C We cannot distinguish depths.
D We cannot distinguish heights.

Explanation:

C: You should try this. Cover one of your eyes with a cloth. Put your hands straight in front of you with the index fingers of both your hands facing each other. Try to bring them close and touch each other very fast. You will find that it is not that easy as compared to when you do it with both your eyes open. That is because the two eyes, together, help in distinguishing different depths. We can tell the difference between two very close objects ie. which one is closer to us, because of both the eyes working together.

Notes